# An Integrated Framework for Identifying Critical Transitions in Single-Cell Trajectories Using HMM, sMCE, and Hodge Decomposition

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## Abstract

We present an integrated computational framework for systematically identifying discrete transcriptional states, critical transition points (tipping points), and directional flow structures from pseudotime-annotated single-cell RNA-seq data. The pipeline integrates four synergistic modules: (1) Hidden Markov Model (HMM) for trajectory discretization with Drug-Resistance Evolution Score (DRES); (2) state-specific Markov Chain Entropy (sMCE) combined with PPI network topology for tipping point detection; (3) Hodge decomposition via Discrete Exterior Calculus (DEC) for resolving transition flows into gradient (irreversible), curl (reversible), and harmonic (plastic) components; and (4) Uncertainty Evolution (UE) index integrating entropy rate, variation of information, and Jensen-Shannon divergence. Applied to drug-tolerance evolution scenarios, our framework reveals fine-grained flow topology that is invisible to conventional trajectory analysis methods.

## Background

A major challenge in both developmental biology and tumor drug-resistance evolution is to identify the critical transition point at which gradual transcriptional variation gives rise to irreversible fate commitment. In the Waddington energy landscape, cells reside in relatively stable attractor states and transition between them through a high-entropy intermediate region, where regulatory instability increases and cell fate becomes especially plastic. Within this transient regime, incremental molecular changes accumulate progressively until they cross a tipping threshold, triggering a qualitative shift in cell identity and pushing the system toward a new stable state. Precisely localizing this turning point is therefore essential for understanding developmental decisions, detecting the onset of drug resistance, and distinguishing plastic intermediate states from fully committed states. Such identification also helps define a potential therapeutic window in which intervention may still redirect cell fate before commitment becomes irreversible.

## Integrated Pipeline Architecture

Our framework processes pseudotime-annotated scRNA-seq data through four integrated stages: HMM discretizes continuous trajectories into transcriptional states; DRES quantifies evolutionary path efficiency; sMCE detects critical tipping points via entropy and PPI topology; Hodge decomposition resolves transition flows into irreversible, reversible, and plastic components; and UE index validates transitions through multi-metric uncertainty convergence.

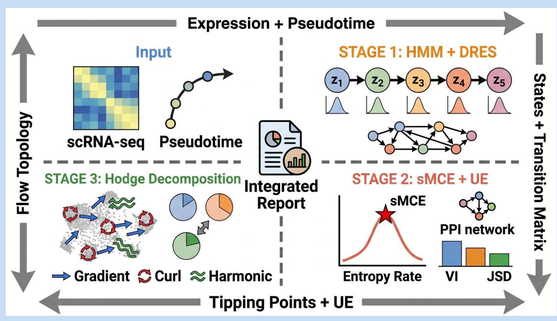


**Figure 1.** The integrated pipeline combines HMM, DRES, sMCE, UE, and Hodge decomposition to identify hidden states, quantify transition efficiency and uncertainty, and resolve the dynamical structure of cell-state transitions from scRNA-seq data.

## Discussion

Single-cell trajectory methods reveal continuous dynamics but lack principled tools to pinpoint when and where critical fate decisions occur along a developmental or evolutionary timeline. Our framework directly addresses this translational need: by integrating **HMM discretization**, **sMCE/UE dual-evidence tipping point detection**, and **Hodge flow decomposition**, it identifies the precise temporal windows in which cells commit to a new state — such as the **transition from drug-sensitive to drug-tolerant in tumor evolution**, or the **irreversible bifurcation point during lineage commitment in development**.

This capability has immediate practical implications. In cancer therapy, identifying the pre-resistance tipping point enables the design of early intervention strategies — administering combination treatments or epigenetic modulators before cells irreversibly commit to resistance, rather than after. The Hodge decomposition further reveals whether a given transition is reversible (high curl — cells still oscillate and may be "pulled back") or irreversible (high gradient — the therapeutic window has closed), providing actionable guidance for choosing between reversal-based and prevention-based strategies. In developmental biology, the same framework localizes the critical stages of lineage specification, informing the optimal timing for reprogramming factor delivery or organoid culture intervention.

Current limitations include the single-trajectory assumption and the heuristic nature of Hodge decomposition on small state spaces. Future integration of multi-modal data (ATAC-seq, spatial transcriptomics) and RNA velocity fields will further refine the temporal resolution and causal interpretability of these critical windows.

## Methodology

Our framework maps continuous pseudotime onto discrete transcriptional states, identifies the critical tipping point where gradual molecular changes trigger fate commitment, and evaluates whether progression toward the terminal state is reversible or irreversible.

### HMM State Discretization

We employ a Gaussian Hidden Markov Model to partition continuous pseudotime trajectories into discrete transcriptional states. Each hidden state captures a distinct transcriptional program, with Gaussian emission distributions modeling the gene expression profiles within that stage. The optimal number of states is determined by BIC /AIC scoring combined with bootstrap resampling to ensure robustness. The resulting state transition matrix reveals the probability of cells moving between states, forming a directed network where node size reflects cell population and edge weight reflects transition likelihood. Building on this, the Drug-Resistance Evolution Score (DRES) evaluates how efficiently cells traverse the sensitive-to-resistant path by scoring forward progress, backward regression, inter-mediate stagnation, and terminal commitment. A higher DRES indicates a more deterministic and irreversible evolutionary trajectory.

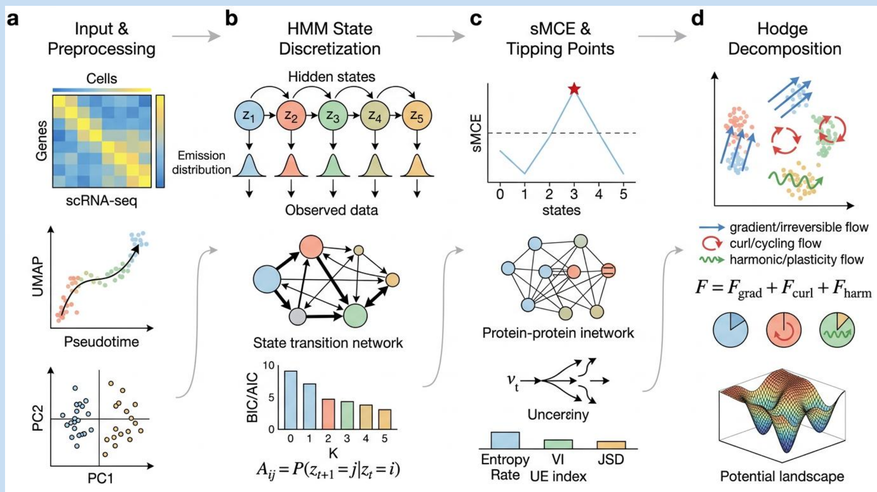


**Figure 2. Overview of the analytical framework for state-transition modeling, tipping-point detection, and flow decomposition.** Starting from scRNA-seq data with pseudotime information, the pipeline performs preprocessing and HMM-based state discretization to construct state transition networks, identifies candidate tipping points using sMCE and uncertainty metrics, and finally applies Hodge decomposition to resolve transition flows into gradient, curl, and harmonic components, thereby linking cell-state dynamics to the underlying potential landscape.

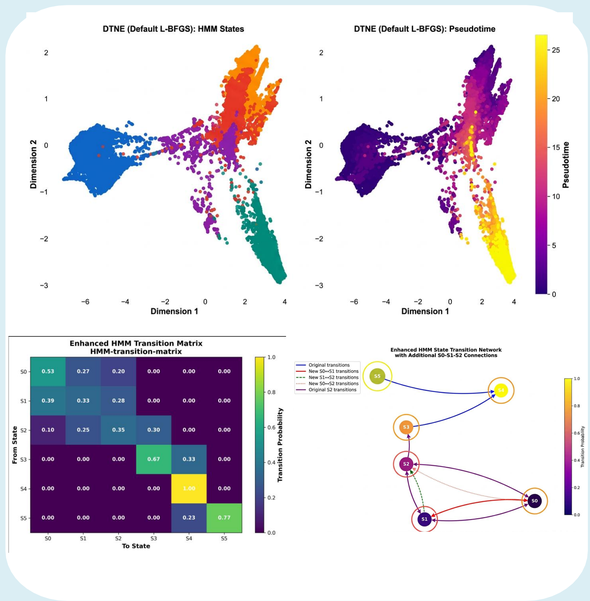


**Figure 3.** HMM state discretization and transition analysis. (Top left) DTNE embedding colored by HMM-assigned states (S0–S5). (Top right) The same embedding colored by pseudotime, showing a continuous gradient from early (purple) to late (yellow) stages. (Bottom left) State transition probability matrix, where S4 acts as a strong absorbing state (P=1.00 from S4→S4). (Bottom right) Enhanced state transition network: node size reflects cell population, edge color and thickness encode transition probability, with dashed lines indicating additional inter-state connections between early states S0–S2.

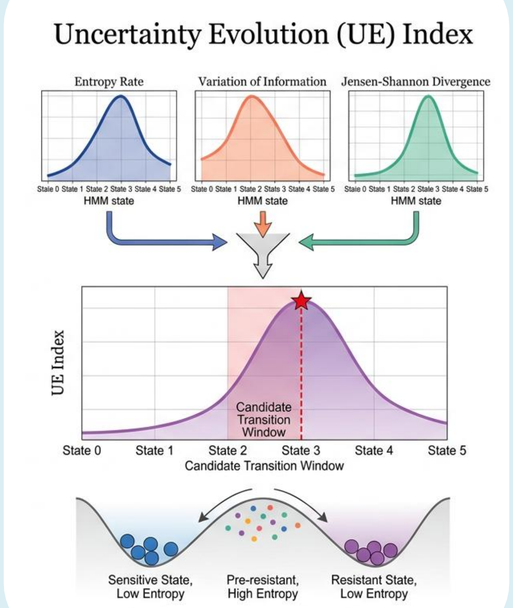


**Figure 5.** The UE Index identifies the pre-resistant State 2-3 interval as the major tipping-point window, where cell-fate uncertainty reaches its maximum between stable sensitive and resistant states.

## Hodge Decomposition of Transition Flows

Inspired by the Helmholtz–Hodge theorem from fluid dynamics, we decompose state transition flows into three orthogonal components, each revealing fundamentally different biological processes:

- **Gradient flow:** Captures irreversible, unidirectional progression along the pseudotime potential — analogous to committed differentiation or drug resistance acquisition.
- **Curl flow:** Identifies reversible, cyclic exchanges between states — associated with cellular heterogeneity and drug-tolerant persister populations.
- **Harmonic flow:** Represents global plasticity cycles (e.g., EMT/MET transitions) that are neither purely gradient nor purely local cycling.

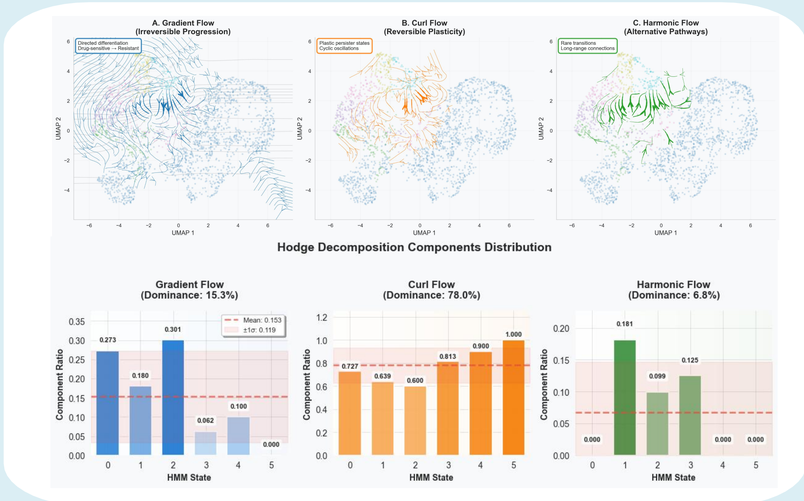


**Figure 4.** Hodge decomposition reveals that state-transition flows are dominated by curl dynamics, with early gradient-driven irreversible commitment and limited intermediate harmonic transitions during the progression from drug-sensitive to resistant states.

## Uncertainty Evolution (UE) Index

The UE index provides multi-dimensional uncertainty quantification by integrating three complementary measures: (i) Entropy rate — quantifying intrinsic randomness of the Markov chain; (ii) Variation of Information (VI) — measuring divergence between adjacent state distributions; (iii) Jensen-Shannon Divergence (JSD) — a symmetric measure of distribution dissimilarity. States where all three metrics simultaneously peak are flagged as high-confidence candidate transition windows.

The framework processes pseudotime-annotatedscRNA-seq data throughfour integrated stages.

HMM discretizes the trajectory into K states.DRES quantifies evolution path efficiency.

sMCE identifies tipping points via entropy. Hodge decomposes flows.

UE provides multi-metric uncertainty evidence.

## Conclusions

(1) Our framework accurately localizes the critical tipping point along pseudotime at which gradual transcriptional variation accumulates to a threshold and triggers irreversible fate commitment, thereby providing a principled strategy for linking continuous cellular trajectories to discrete state transitions in the Waddington landscape and for resolving the boundary between stable attractor states, transitional high-entropy regions, and terminal committed states. This enables a more interpretable description of how cell populations progress through unstable intermediate regimes before entering a new stable fate, and offers a quantitative way to define the moment at which qualitative biological change emerges from continuous molecular variation.

(2) By distinguishing plastic intermediate states from fully committed terminal states and quantitatively assessing whether progression toward the final state remains reversible or has become irreversible, the framework improves the interpretation of both developmental processes and tumor drug-resistance evolution. It further clarifies the timing, directionality, and stability of state transitions, helps reveal when cells remain susceptible to perturbation, and defines a biologically meaningful window for early intervention before cell fate is permanently fixed.

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